

teamscale-mcp & teamscale-docs-mcp

gitlab.com/cqse/internal/teamscale-mcp · **teamscale-docs-mcp**

Two sidecars: the full REST API plus the Teamscale docs, for any client

⚠ Prototype

This is only a **prototype**

*Built to demonstrate the **value** of the MCPs, not production-ready code.*

Expect rough edges. The point is to show what these MCPs make possible, not to ship a finished product.

Three pieces, three jobs

-  **Teamscale Claude Code plugin**: curated skills for developers editing code
-  **teamscale-mcp**: the full REST API as tools (the data)
-  **teamscale-docs-mcp**: the Teamscale docs as tools (the understanding)

The plugin helps a developer change the code Teamscale analyses. These two help us, and the customers we work with, read, explain, and configure Teamscale itself.

This talk is about that second pair.

Quick recap: published plugin

The official `teamscale/claude-code` plugin:

- Per-user and Claude-Code-only, installed on each workstation
- Bundles curated skills (`fix-findings` , `pr-*` , `local-*`) ...
- ... plus a local backend (`ts_agent_helper`) that those skills call
- Needs Python, the `teamscale-dev` CLI, a `.teamscale.toml` , and a local checkout

Its value is **opinionated workflows for a developer fixing findings** in the analysed code, with the repo checked out on their own machine.

What **teamscale-mcp** is

One central HTTP sidecar, deployed once next to Teamscale:

- *One tool per REST endpoint*, generated at startup from the instance's live OpenAPI spec
- *Per-request identity* via `X-Teamscale-User` / `-Token` headers
- Reachable by *any MCP client* over a URL

If the API can do it, there's a tool for it, and it always matches the instance's own version.

Configured entirely by **env vars**

teamscale-mcp:

```
image: registry.gitlab.com/cqse/internal/teamscale-mcp:latest
ports: ["8081:8081"] # serving: MCP_HOST / _PORT / _PATH, MCP_ALLOWED_HOSTS
environment:

# Connection + bootstrap creds (used once, at startup, to fetch the spec)
TEAMSACLE_SERVER_URL: http://teamscale:8080
TEAMSACLE_SPEC_USER: some-technical-user
TEAMSACLE_SPEC_TOKEN: some-technical-user-api-token
TEAMSACLE_INCLUDE_INTERNAL: "false"

# Tool surface: trim to a sharp, read-oriented set (blocklist recommended)
TEAMSACLE_EXCLUDE_TAGS: "Backup,System,Users,Profilers,SAP,..."
TEAMSACLE_EXCLUDE_METHODS: "DELETE,PUT,PATCH"
# TEAMSACLE_INCLUDE_TAGS: "Findings,Metrics,Test Gap Analysis"
# TEAMSACLE_INCLUDE_NAMES: "createBaseline"
```

Per-request identity arrives in the `X-Teamscale-User` / `-Token` headers, never from the environment.

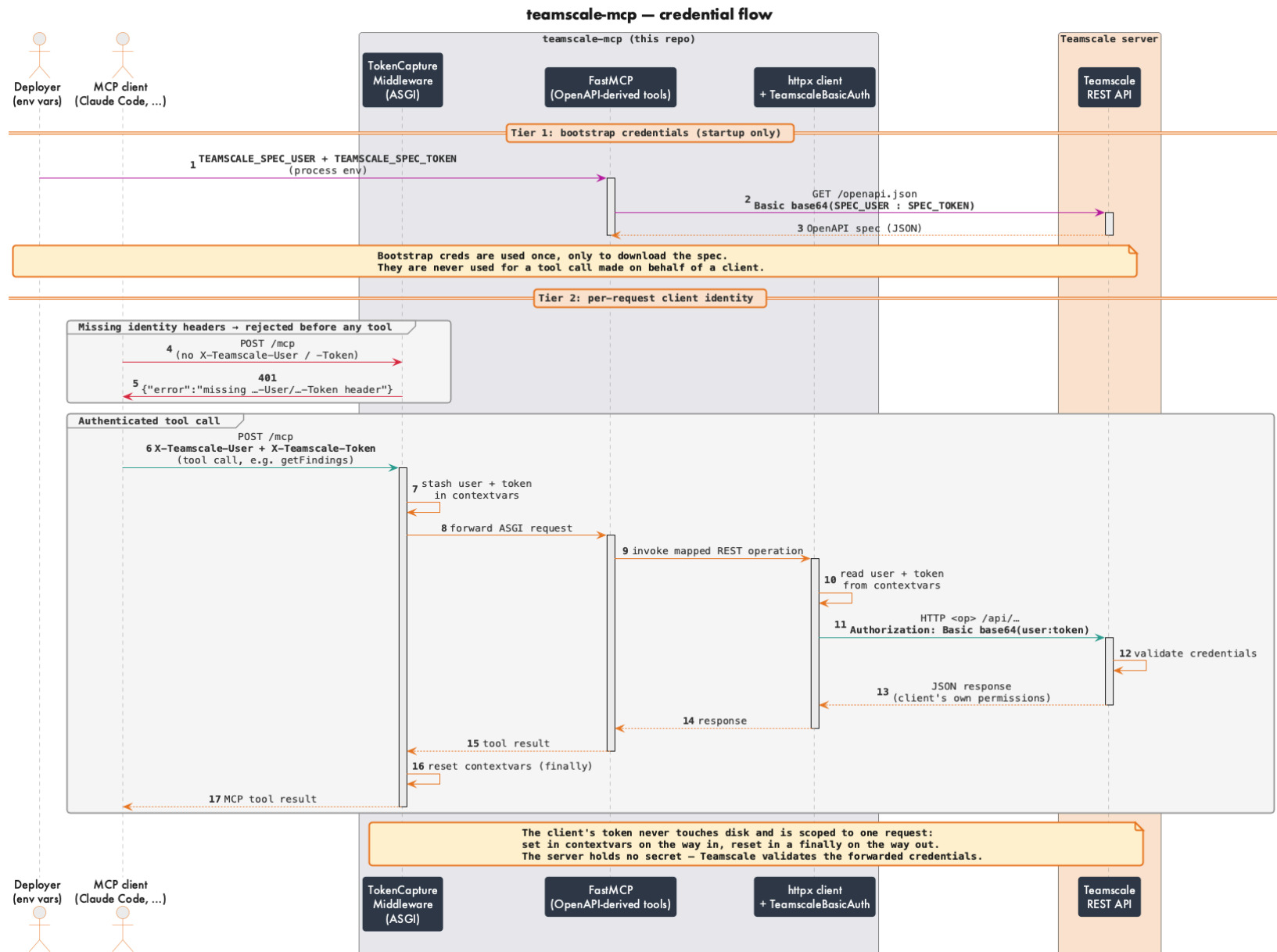
Connect a client with **one command**

Point any MCP client at the URL, presenting your own identity:

```
claude mcp add teamscale-cqse --scope user --transport http \  
  https://cqse.teamscale.io/mcp \  
  --header "X-Teamscale-User: bruckner@cqse.eu" \  
  --header "X-Teamscale-Token: YOUR_API_TOKEN"
```

The two headers *are* your Teamscale identity, forwarded per request.
Any MCP-capable client works the same way.

The credential flow: two tiers, no stored secret



The honest reframe: what it is *not*

It is **not** a drop-in replacement for `ts_agent_helper`:

- Only 3 of 6 skills route through it
- `fix-findings` is a clean win (pure REST)
- `pr-*` is a wash, because the agent has to redo the git/PR resolution the CLI bundled
- `local-*` can't move at all, since they need the local working tree

"Replaces `ts_agent_helper`" is the weakest way to sell it. The next slides are the strong way.

Where **teamscale-mcp** really fits

The real value is reach the per-user plugin can't offer:

- 🔍 **Data access** across the entire API, for questions no skill encodes
- 🏢 **One central deployment** per instance with no per-workstation Python
- 📈 **Observability**: one server, and every tool call is routed through it
- 🤖 **CI and service accounts**: an HTTP endpoint any client can call, as a technical account rather than a person's login
- 🧠 **Any LLM integration**: plain MCP over HTTP, so it's not tied to Claude Code; any MCP-capable client or model can use it
- 🧑 **Weakest fit**: an individual dev's daily fix loop, where you'd just use both

A different **audience**, not just a different deployment

The sharpest distinction isn't *how* it's run, it's *who* it's for.

	Plugin	teamscale-mcp (+ docs)
Works on	the analysed code	the Teamscale data & config
Assumes	a checkout in front of you	just a URL and an identity
Built for	developers editing code	people understanding Teamscale

- Us on customer instances
 - The customer managers we report to
 - and newcomers setting an instance up
- People who read Teamscale's results and act on them, without shipping the analysed code.

teamscale-docs-mcp, the understanding half

Teamscale's documentation, served as MCP tools:

- A *sitemap catalog*, plus individual pages converted to Markdown on demand
- *No auth*, because the docs are public content
- Points at an instance's *version-matched bundled docs*, or at the public site
- Self-hosted alongside Teamscale, or run as one central endpoint

It hands the agent Teamscale's manual, not just its data.

Configured by a single **env var**

```
teamscale-docs-mcp:
```

```
image: registry.gitlab.com/cqse/internal/teamscale-docs-mcp:latest
ports: ["8082:8082"]           # serving: MCP_HOST / _PORT / _PATH, MCP_ALLOWED_HOSTS
environment:

# The only real setting: where the docs live. Point at the instance's
# version-matched bundled docs, so no request leaves the network ...
DOCS_BASE_URL: http://teamscale:8080/documentation

# ... or serve the public site instead:
# DOCS_BASE_URL: https://docs.teamscale.com
```

No credentials, no headers: the docs are public content, so there is nothing to authenticate.

Explain and act: generic docs, tailored to *your* repo

Neither half is the point on its own. Together they turn a generic manual into steps for *this* repo.

- 📖 **teamscale-docs-mcp** explains: the generic how-to (any language, any stack)
- 🔌 **teamscale-mcp** acts: *this* project's real languages, test framework, build/CI config
- 📝 Claude merges the two into a runbook written for this repo specifically

Same how-to page, different repo → a different, correct set of steps every time, not a generic doc dump the reader has to adapt themselves.

What "explanation" actually looks like

-  **A customer's manager:** *"This test gap is 34%. Is that bad, and where did it come from?"*
 The docs explain the concept, and teamscale-mcp pulls the number and trend.
-  **New to Teamscale:** *"How do I set up an analysis profile for Kotlin?"*
 The docs give the how-to, and teamscale-mcp creates and wires it.
-  **Us, on a customer instance:** *"Compare clone coverage across these five projects."*
 Full REST reach, with the reference at hand, across customer instances.
-  **Making sense of a codebase:** *"What's the architecture of my system?"*
 teamscale-mcp queries the code, drafts a Teamscale **architecture**, and sets it up in Teamscale to analyze against going forward.

Understand first, act second, always grounded in the customer's real data.

A real session: TGA + frontend profiling

Not a mockup, but an actual conversation against a live project:

- Learn what Test Gap Analysis *is* (docs)
- Pull its real number (teamscale)
- Get a setup runbook for frontend coverage (docs), tailored to this project's stack

Docs and data interleaved end to end.

I want to know about TGA in Teamscale

• I'll look this up in the Teamscale documentation rather than rely on memory. Let me load the documentation tools.

Called teamscale-marcelbruckner-com-docs 4 times

• TGA = Test Gap Analysis – one of Teamscale's core "Test Intelligence" features. Here's what it is and how it works.

The problem it solves

Most production bugs occur in **recently changed code**. But testers often don't know everything that changed – refactorings, undocumented bug fixes, or whole features get shipped without the test team being told. The result: risky changes go to production untested. (CQSE's own research found this accounts for the majority of production defects.)

What TGA does

Test Gap Analysis combines three things Teamscale already tracks – **code changes**, **test coverage**, and **issues** – to visualize *every* recently changed method and whether it has been tested yet.

The signature view is the **Test Gap treemap**, where each rectangle is a changed method:

- Gray – unchanged code
- Green – changed *and* tested
- Red / orange – changed but **never tested** → these are your *Test Gaps*

This lets testers spot untested risk and schedule additional test cases *before* release.

Key capabilities

- Works across all **test stages**, not just unit tests: integration, end-to-end, manual, and even hardware/software-in-the-loop tests. Many teams see the true coverage of these non-unit stages for the first time.
- **Issue-level TGA** – aggregates test gaps per Jira issue / Azure DevOps work item, so a tester can see exactly how much of the code changed for a given story/feature is still untested, and filter/prioritize accordingly.
- **Pull-request integration** – Teamscale can post PR comments flagging open test gaps, putting the info directly in front of developers.
- **KPIs & trends** – long-term coverage trend per stage, unaddressed gaps per PR, coverage per user story/defect/epic. Supports a "shift-left" feedback loop.

Getting started (relevant docs)

Task	Page
Record coverage via the profiler	/howto/setting-up-profiler-tga/ (Java, C#, C++, JS, Node, Python...)
Coverage for manual tests	/howto/setting-up-profiler-tga/manual-tests/
Reading the treemaps	/howto/working-with-test-gap-treemaps/
Issue-level test gaps	/howto/analyzing-issue-test-gaps/
Build a TGA dashboard	/howto/creating-a-tga-dashboard/

| Note: TGA (Test Gap Analysis) is distinct from Teamscale's smart **test suggestions** (Test Impact Analysis / Pareto) – those focus on which tests to run fast, whereas TGA focuses on which changes are untested.

Want me to walk through any specific part – e.g. setting up the profiler to record coverage, reading the treemap, or issue-level test gaps?

* Cogitated for 37s

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A real session: TGA + frontend profiling · 2. pull the number (first try)

I want to know about TGA in my traureise project

Called teamscale-marcelbruckner-com

- Found three matching projects: traureise, traureise-fastapi-react, and traureise-original. Let me pull the TGA data for the exact traureise project.

Called teamscale-marcelbruckner-com 2 times

- Here's what I found for your Traureise project (traureise):

TGA result

Test Gap percentage: 0%

- Project: Traureise (traureise), default branch main, configuration completed.
- Computed over the full default-branch history (default baseline, all default coverage partitions).

How to read this

A TGA percentage of 0% means zero percent of the analyzed changed code is showing up as a test gap. There are two very different reasons this can happen, and it's worth confirming which one applies to you:

1. Everything relevant is covered – all changed methods in scope have been executed by tests. Genuinely 0 gaps. 🏆
2. No test coverage has been uploaded – with no coverage data, there's nothing to compare changes against, so the metric collapses to 0. This is the more common cause for a newer project.

Given the project was created fairly recently and I can't see any coverage partitions referenced, cause #2 (no coverage uploaded yet) is the likely explanation – but I'd want to verify rather than assume.

Heads up: three matching projects

There are actually three projects with "traureise" in the name:

- traureise ← this one
- traureise-fastapi-react
- traureise-original

If you meant a different one, tell me and I'll pull its numbers.

Want me to dig deeper?

I can:

- Check whether coverage exists (query with all-partitions / execution-only to see if any methods were executed) to confirm which of the two cases above applies,
- Break down the Test Gap treemap to see changed vs. tested vs. untested methods,
- Pull issue-level test gaps for specific tickets, or
- Run the same for traureise-fastapi-react / traureise-original.

Which would you like?

No, I'm sure coverage exists

- Good – then let me figure out why the default query returned 0%. My first call used only the default coverage partitions; if your coverage lives in other partitions it would be ignored. Let me re-run with all partitions considered, and separately check whether methods are being seen as executed at all.

Called teamscale-marcelbruckner-com 3 times

- All queries – including the churn-only one, which ignores coverage entirely – return 0. That's the key clue: churn TGA of 0 means Teamscale sees no *changed* methods in the analyzed window. TGA always measures "of the code that changed since a baseline, how much is untested" – so with no baseline set, the change window is effectively empty and the result collapses to 0 regardless of coverage.

Let me check the configured baselines and re-run over an explicit time window (from before the project's first commit through HEAD).

Called teamscale-marcelbruckner-com 2 times

- That's the answer. The reason every earlier query returned 0% is now clear:

Root cause

Your project has no configured baselines (`getAllBaselines` → empty). TGA measures test gaps *relative to a baseline* (the start of the change window). With no baseline set, the default query had an empty change window → 0 changed methods → 0% by definition. It was never a coverage problem – your coverage is fine, as you said.

Actual TGA for Traureise

Measuring across the full project history (from before the first commit through `main:HEAD`, all coverage partitions):

Test Gap ratio: ~22.1%

Interpreting the treemap semantics:

- ~22% of the methods changed since the project began have never been executed by a test – these are your test gaps.
- ~78% of changed methods have been covered. 🙌

Why this matters for your setup

Because there's no baseline defined, the TGA widgets/dashboards in the UI will also show 0 by default until you give them a time window. Two fixes:

1. Configure a baseline (e.g., "last release" or a fixed date) under project settings → then TGA reflects gaps *since that point*, which is the usual way teams track "did we test what we changed this release." See </howto/working-with-test-gap-treemaps/>.
2. For ad-hoc analysis, just specify a start/end time window (as I did here) rather than relying on the default.

Want me to:

- Break down the treemap to show which specific methods/files make up that 22% gap,
- Compute TGA since a specific date (e.g., a recent release) instead of all-history, or
- Check issue-level test gaps for particular tickets?

● I've localized the 22% gap down to the file level – that's as granular as the API goes (per-method red/orange cells only exist in the UI treemap). Here's the breakdown.

Where the ~22% test gap lives

Frontend – the bulk of the gap

Path	Test gap
frontend/src (entire tree)	100%

None of your frontend source (components/, routes/, hooks/, etc.) maps to any recorded coverage. Even though there are Playwright specs under frontend/tests, no coverage from them is being associated with the frontend source methods in Teamscale – so every changed frontend method counts as a gap. This is the single largest contributor.

Backend – small, concentrated gap (backend/app = 13%)

File	Test gap	Note
backend/app/api/routes/utils.py	100%	fully untested
backend/app/utils.py	14.3%	~1 in 7 changed methods untested
backend/app/api/routes/login.py	20%	e.g. password-recovery / reset handlers typically
backend/app/crud.py	0%	fully covered ✓
backend/app/core/*	0%	✓
backend/app/api/routes/items.py, users.py, private.py, websites.py	0%	✓

Reading this

- Your backend is in good shape – gaps are limited to utility/helper code (utils.py, the routes/ utils.py endpoints) and part of login.py. Those are the concrete methods to add tests for.
- The headline number is dominated by the frontend, which shows as 100% gap. Before writing a pile of frontend tests, it's worth checking why – most likely the frontend coverage (Playwright/Vitest) either isn't being uploaded or isn't mapped to a coverage partition Teamscale associates with frontend/src. If you expected frontend coverage to exist, that's a coverage-upload/mapping issue rather than a genuine testing gap.

Getting exact method names

The public API exposes TGA only as a percentage per path, not the individual red/orange methods. To see the exact untested methods, open the Test Gap treemap in the Teamscale UI for traureise (set the same time range / a baseline) – each red or orange cell is a specific changed-but-untested method, and clicking it jumps to the code.

Want me to:

- Drill the frontend into subfolders (components vs routes vs hooks) to rank them, or
- Check whether any frontend coverage partition exists at all (to confirm the upload/mapping theory)?

Can you help me setting up coverage profiling for frontend code?

- This is a documented setup path, so let me pull Teamscale's own frontend-coverage guides before advising – I want to give you the exact steps for your stack (Vite + React + TypeScript with Playwright e2e tests), not generic ones.
- Wrong tool name – the docs server has its own prefix. Let me use the correct one.

Called teamscale-marcelbruckner-com-docs 2 times

- I've got Teamscale's frontend-coverage guide. Your frontend is Vite + React + TypeScript with Playwright e2e tests (frontend/tests/*.spec.ts, playwright.config.ts) – that's exactly the scenario the JavaScript profiler is built for: an app served to a browser, tested by automated UI tests.

Before I write you the concrete runbook, two decisions materially change the steps – let me pin them down.

- User answered Claude's questions:

- └ · Which coverage-collection approach do you want to use for the frontend? → Teamscale JS Profiler (Recommended)
- Which test stage(s) should produce the frontend coverage that feeds TGA? → Playwright E2E, Unit tests (Vitest/Jest)
- Do you have the traipse frontend checked out locally so I can edit config files, or should I give you a written runbook? → Written runbook

A real session: TGA + frontend profiling · 6. the runbook (setup)

• Here's your runbook. Since you want **both** stages, note one important split up front:

- Playwright E2E runs in a real browser → use the **Teamscale JavaScript Profiler** (instrumenter + collector). This is the recommended path.
- Vitest unit tests run in Node/jsdom → the profiler *can't* be used there (instrumented code fails in Node). Vitest produces its own coverage, which you upload separately.

So you'll end up with **two coverage partitions** feeding TGA, e.g. E2E Coverage and Unit Test Coverage.

Part A – One-time setup in Teamscale (both stages)

1. Go to **Project Configuration** → **Coverage Profilers** in the traureise project.
2. Create a profiler config, ID `traureise-e2e`:
teamscaleProject=traureise
teamscalePartition=E2E Coverage
dumpAfterMins=120
2. (The Vitest path doesn't need a profiler config – it uploads directly via teamscale-upload.)

Part B – Playwright E2E coverage (JS Profiler)

1. Enable source maps (so `dist/` maps back to `frontend/src`)

```
frontend/tsconfig.json:
{ "compilerOptions": { "sourceMap": true, "inlineSources": true } }
frontend/vite.config.ts:
export default defineConfig({
  build: { sourcemap: true },
})
```

2. Build and instrument the app

The profiler instruments the **built** output – you must serve the built+instrumented app, *not* the Vite dev server.

```
cd frontend
npm run build                # produces ./dist with source maps
```

```
npx @teamscale/javascript-instrumenter \
  --input ./dist/ \
  --collector localhost:54678 \
  --config-id traureise-e2e \
  --commit "$(git rev-parse HEAD)" \
  --in-place \
  --include-origin 'src/**/*'
- --commit ties the coverage to the exact revision being tested (essential for TGA).
- --include-origin 'src/**/*' matches your frontend/src sources.
```

3. Start the collector (auto-upload to Teamscale)

Add to `frontend/package.json`:

```
"scripts": {
  "collector": "coverage-collector --teamscale-server-url https://teamscale.marcelbruckner.com --teamscale-user <user> --teamscale-access-key <key> --teamscale-project traureise",
  "pree2e:cov": "npx pm2 delete CC; npx pm2 start npm --name CC -- run collector",
  "e2e:cov": "playwright test",
  "poste2e:cov": "npx pm2 delete CC"
}
```

Install: `npm install -D @teamscale/coverage-collector pm2`

4. Point Playwright at the instrumented build

In playwright.config.ts, make webServer serve the built dist/ (e.g. vite preview or npm run preview), not vite dev:
webServer: { command: 'npm run preview', url: 'http://localhost:4173', reuseExistingServer: false }

5. Content Security Policy (only if you set one)

The FastAPI full-stack template doesn't ship a strict CSP by default, so the WebSocket to the collector should just work. If you later add a CSP, it must allow the collector:
connect-src 'self' ws://localhost:54678;
script-src 'self' blob: ws://localhost:54678;
worker-src 'self' blob: ws://localhost:54678;

6. Run it

npm run e2e:cov
The collector receives coverage over WebSockets as Playwright drives the browser, then uploads to the E2E Coverage partition.

Part C – Vitest unit-test coverage

| If you don't have a Vitest suite yet, this stage produces nothing until you add one. Say the word and I'll help scaffold it.

1. Enable coverage in Vitest

```
cd frontend
npm install -D @vitest/coverage-istanbul
vite.config.ts (or vitest.config.ts):
test: {
  coverage: {
    provider: 'istanbul',
    reporter: ['lcov'],           // lcov → Teamscale reads it directly
    reportsDirectory: './coverage',
    include: ['src/**/*.ts'],
  },
}
```

2. Run and upload

```
npx vitest run --coverage
npx @cqse/teamscale-upload \
  --server https://teamscale.marcelbruckner.com \
  --project traureise \
  --user <user> --accesskey <key> \
  --partition "Unit Test Coverage" \
  --format LCOV \
  --commit "$(git rev-parse HEAD)" \
  ./coverage/lcov.info
```

Part D – Verify

After each upload, TGA for the frontend should stop reading 100%. You can confirm by asking me to re-run getTgaPercentage for frontend/src (all partitions), or check the Test Gap treemap in the UI.

A couple of things worth flagging:

- The instrumenter must run against the built bundle each time you test a new commit – wire steps B2-B6 into your CI so coverage stays current, otherwise TGA drifts stale.
- Use git rev-parse HEAD consistently for --commit in both stages so E2E and unit coverage land on the same revision.

Want me to (a) generate a ready-to-paste CI job (GitHub Actions, matching your repo layout), or (b) help scaffold the Vitest suite if you don't have one yet?

The same pairing, on a tool outside either server

`teamscale-docs-mcp` also documents `teamscale-dev` .

`teamscale-dev` itself stays out of scope for both servers, but *explaining* it doesn't.

The same pairing, on a tool outside either server

Check the teamscale docs. How do i configure teamscale-dev

- I'll search the Teamscale docs for how to configure teamscale-dev.

Called teamscale-marcelbruckner-com-docs 2 times

- Found it: `/reference/cli/teamscale-dev/`. Let me read that page.

Called teamscale-marcelbruckner-com-docs

- Let me also pull the `.teamscale.toml` reference, since that's the config file teamscale-dev reads.

Called teamscale-marcelbruckner-com-docs

- Here's how to configure teamscale-dev (the Teamscale developer CLI). There are two distinct pieces: **what to analyze** (a `.teamscale.toml` file) and **how to authenticate** (server credentials).

1. Project config: `.teamscale.toml`

teamscale-dev reads `.teamscale.toml` files (TOML format) to know which server/project/branch a directory maps to. A minimal file:

```
server.url = "https://teamscale.io/"
project.id = "example"
```

Full set of properties:

```
version = "1.0" # optional, default 1.0
root = false   # if true, config discovery stops here and doesn't look further up
```

```
[server]
url = "https://teamscale.io/" # must resolve in the merged config; must match a server you have credentials for
```

```
[project]
id = "example" # the Project ID (may differ from the display name)
branch = "main" # optional; if unset, discovered from VCS, falling back to server default
path = "remote/path" # optional; path within the Teamscale project (" " = project root)
```

Notes:

- **Merging:** files inherit from parent directories (like `.editorconfig`). Put `server.url` + `project.id` in your repo root, and override just `project.path` in subdirectories. A `root = true` file stops upward discovery.
- **Auto-generate it:** in Teamscale, go to *Project Configuration* → *Projects* and click *Configure IDE plug-ins / CLI* to download a ready-made file.

The same pairing, on a tool outside either server

2. Credentials (username + access key)

Config files hold no secrets – you supply your username and access key separately. Five ways, in precedence order:

Method	How
--server option	--server https://user:accesskey@example.com/teamscale/ (can repeat per server)
TEAMSCALE_DEV_SERVERS env var	Same credentials-URL format, space-separated for multiple servers
--user / --accesskey options	--user NAME --accesskey KEY (applies to all servers not matched by --server)
TEAMSCALE_DEV_USER / TEAMSCALE_DEV_ACCESSKEY env vars	Env equivalent of the above
Args file	Put --server ... in ~/.teamscale-dev.args, then pass @"\${HOME}/.teamscale-dev.args"

Recommended for terminal use – set once in your ~/.zshrc (you're on zsh/macOS):

```
export TEAMSCALE_DEV_SERVERS="https://username:accesskey@example.com/teamscale/"
```

Then you never need credential flags. ⚠️ Avoid passing --accesskey/--server directly on the command line in shared environments – other users can read process args (ps). Prefer env vars or the args file. Note @ in a username must be percent-escaped as %40 (though teamscale-dev auto-corrects @).

Verify it works

```
teamscale-dev verify-configuration src/ # validates .teamscale.toml + server/project access
```

Common commands once configured

- teamscale-dev pre-commit src/ – upload local changes for pre-commit analysis and show findings (add --upload-uncommitted-files when paths are under Git/SVN – recommended)
- teamscale-dev fetch-findings src/ – show findings already on the server (no upload)
- teamscale-dev open-in-browser src/ – open the file/dir in the Teamscale web UI
- teamscale-dev mcp – run as an MCP server for AI coding assistants
- teamscale-dev generate-completion – shell tab-completion (bash/zsh)

Other useful flags: --branch (override branch), --insecure or a #trust-all-certificates URL suffix (disable TLS validation – testing only), --debug/--trace (logging).

Want me to help you write a .teamscale.toml for a specific project or set up the zsh env vars?

The same pairing, on a tool outside either server

and why do i need user and accesskey then

- Good question – you don't need both. They're alternative ways to supply the same credentials, useful in different situations.

The distinction is how credentials get matched to a server URL:

- TEAMSCALE_DEV_SERVERS (and the --server flag) – credentials are bound to a specific server URL, because the username and access key are baked into the URL itself (https://user:key@example.com/...). This is per-server.
- TEAMSCALE_DEV_USER / TEAMSCALE_DEV_ACCESSKEY (and --user / --accesskey) – credentials that are not tied to any particular URL. They apply as a fallback to every server referenced in your .teamscale.toml files that doesn't already have credentials from a --server/TEAMSCALE_DEV_SERVERS entry.

So:

Situation	What to use
One server, or same login everywhere	Either one alone is enough – TEAMSCALE_DEV_SERVERS is fine
Different logins for different servers	TEAMSCALE_DEV_SERVERS (each URL carries its own credentials)
One default login for whatever server the config points at	TEAMSCALE_DEV_USER + TEAMSCALE_DEV_ACCESSKEY

If you've set TEAMSCALE_DEV_SERVERS for your server, you do not also need TEAMSCALE_DEV_USER/TEAMSCALE_DEV_ACCESSKEY. The --server form takes precedence anyway.

Why offer the --user/--accesskey form at all? Because it separates the URL (which lives in the shared, committed .teamscale.toml) from the credentials (which are personal). With it you can keep just server.url = "https://example.com/teamscale/" in the project config and let each developer supply their own user/key via env vars – no secrets in the URL, no secrets in version control.

So for your single-server case: pick one approach. TEAMSCALE_DEV_SERVERS alone is all you need.

The honest ledger: **capability, not opinion**

✓ **Can**

- Expose the **entire REST API** as tools
- Serve **any MCP client** (Claude Code, CI, web, cron)
- Authenticate as **any identity** (personal or service)
- **Explain and act** together (with docs-mcp)
- **Observability**: every call through one vantage point
- **Retrofit** old instances

✗ **Can't**

- **No local working tree** → `local-*` stay on `teamscale-dev`
- **Raw tools, not workflows** → orchestration on the agent, so not the smoothest fix/PR loop

It's an addition, not a replacement for `ts_agent_helper`

Future work: from **prototype** to product

Today — a prototype

- A **working proof of value**, not a finished product
- Shows how much an agent gains with Teamscale in reach: **query and act** (`teamscale-mcp`) and **understand in context** (`teamscale-docs-mcp`)
- Runs as a **sidecar**, bolted on from the outside

Next — part of Teamscale

- The value is proven; the **home is inside the product**
- A **first-class, native** MCP capability
- **Shipped, versioned, and supported** with Teamscale

We see a lot of **value for our customers** here, and we'd love to see this become a part of Teamscale.

Thank you

teamscale-mcp gives Teamscale's data to any agent.

teamscale-docs-mcp gives it the manual.

Questions?



Backup

Connect multiple clients with **one command each**

Point any MCP client at the URL, presenting your own identity:

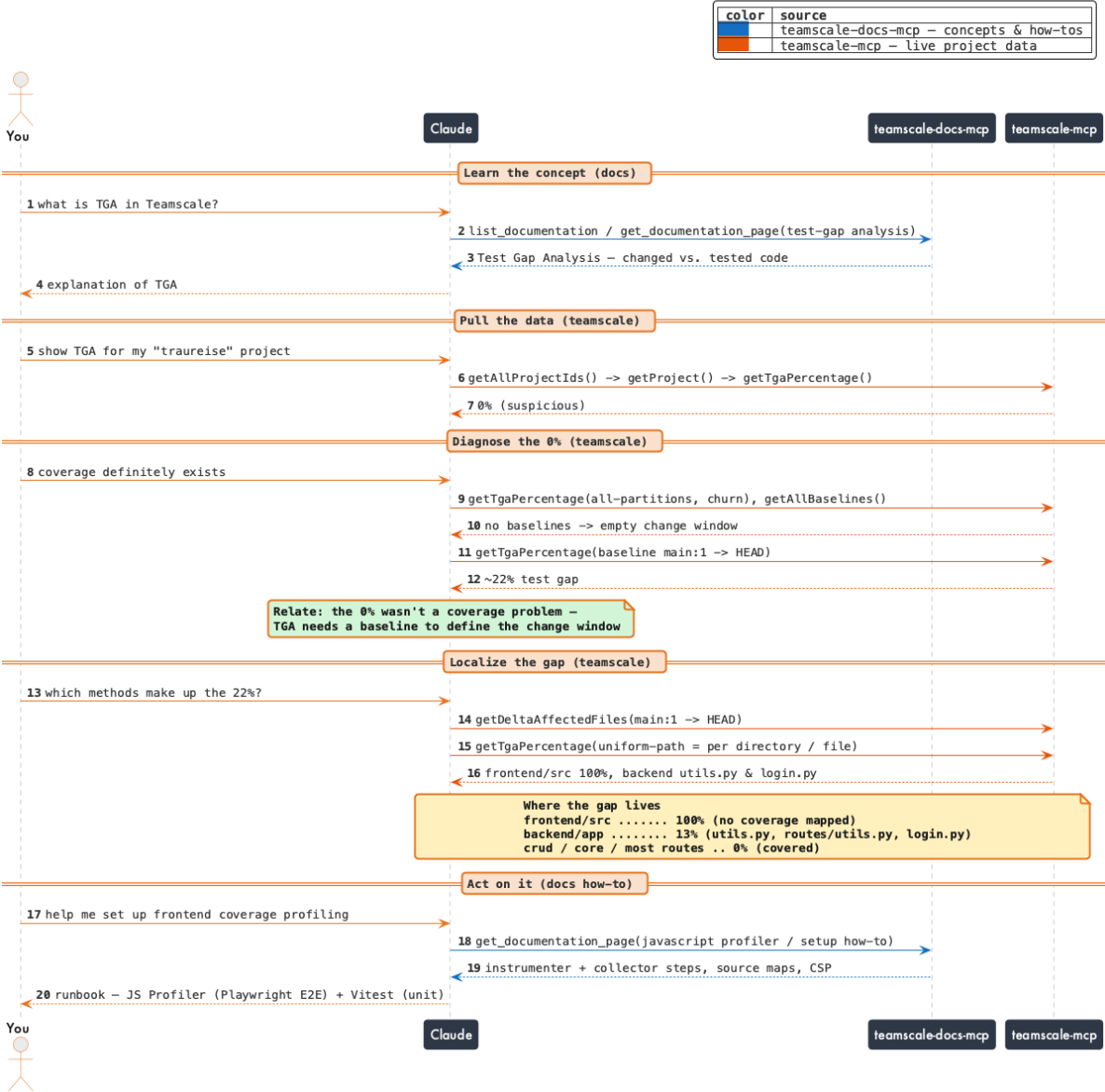
```
claude mcp add teamscale-cqse --scope user --transport http \
  https://cqse.teamscale.io/mcp \
  --header "X-Teamscale-User: bruckner@cqse.eu" \
  --header "X-Teamscale-Token: YOUR_API_TOKEN"
```

And add as many servers as you need:

```
claude mcp add teamscale-personal --scope user --transport http \
  https://teamscale.marcelbruckner.com/mcp \
  --header "X-Teamscale-User: marcel" \
  --header "X-Teamscale-Token: YOUR_API_TOKEN"
```

The two headers *are* your Teamscale identity, forwarded per request.
Any MCP-capable client works the same way.

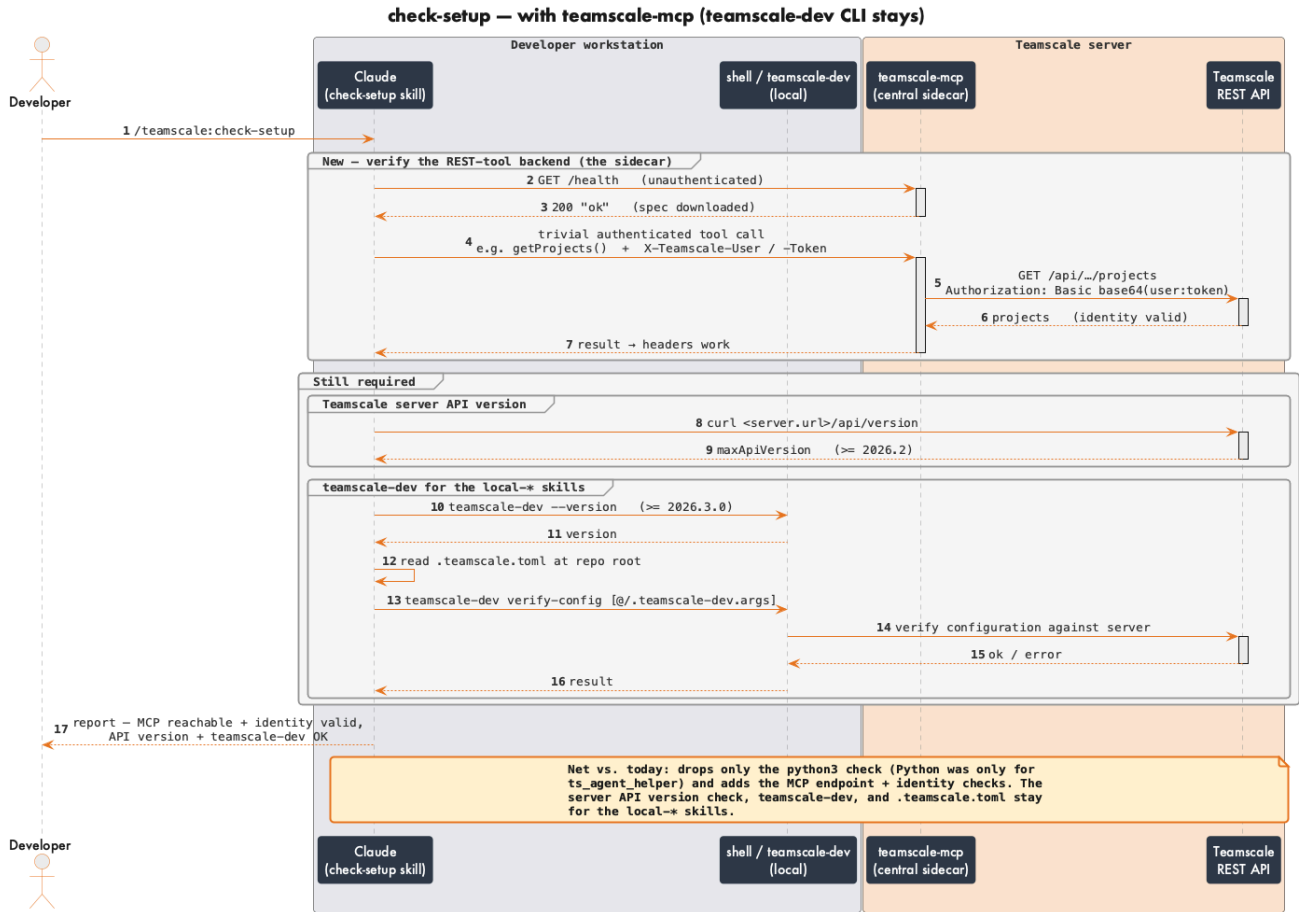
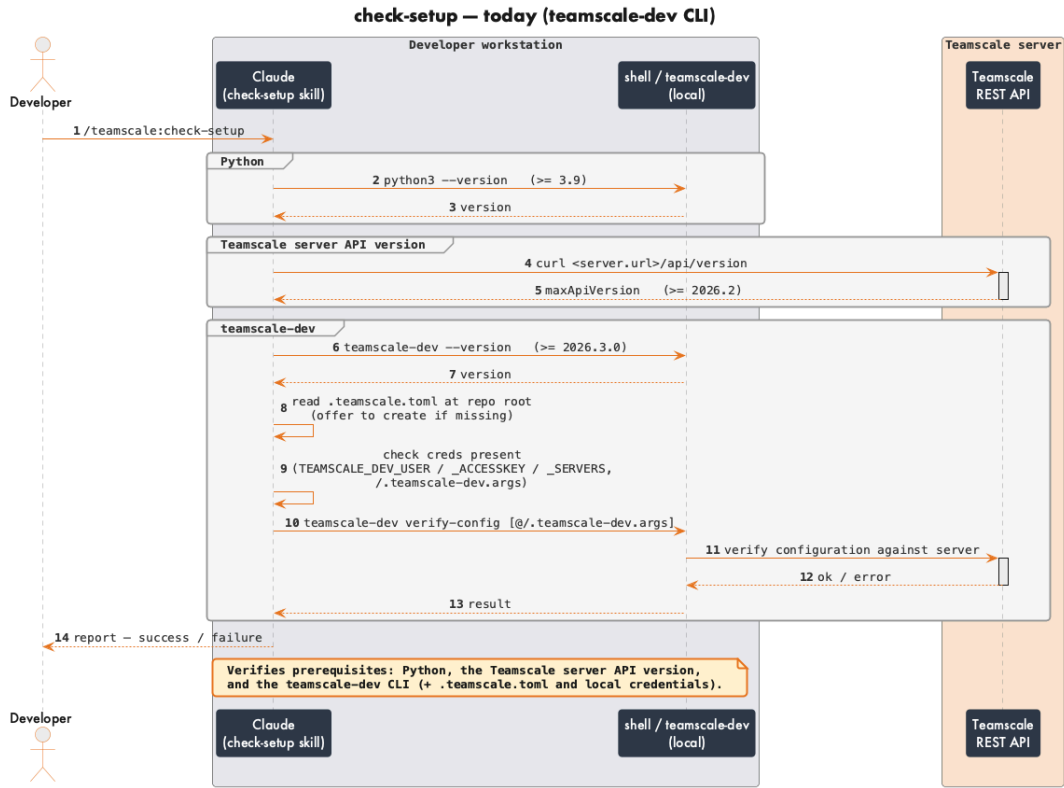
Check TGA & set up frontend profiling — you, teamscale-mcp & teamscale-docs-mcp

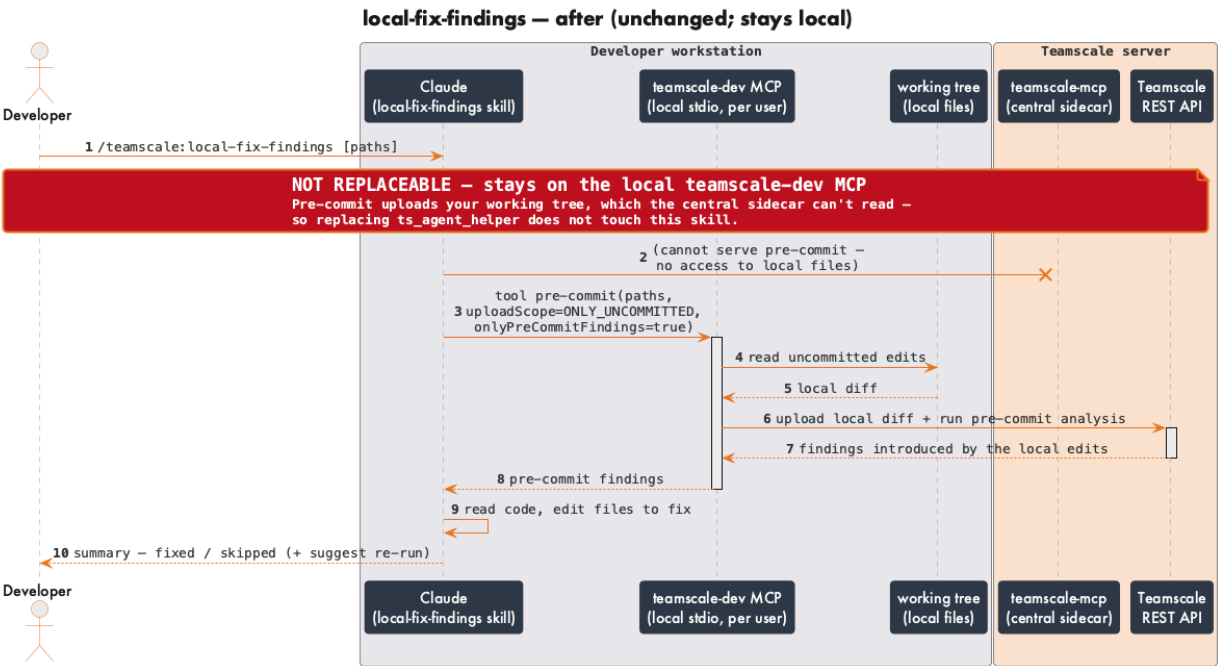
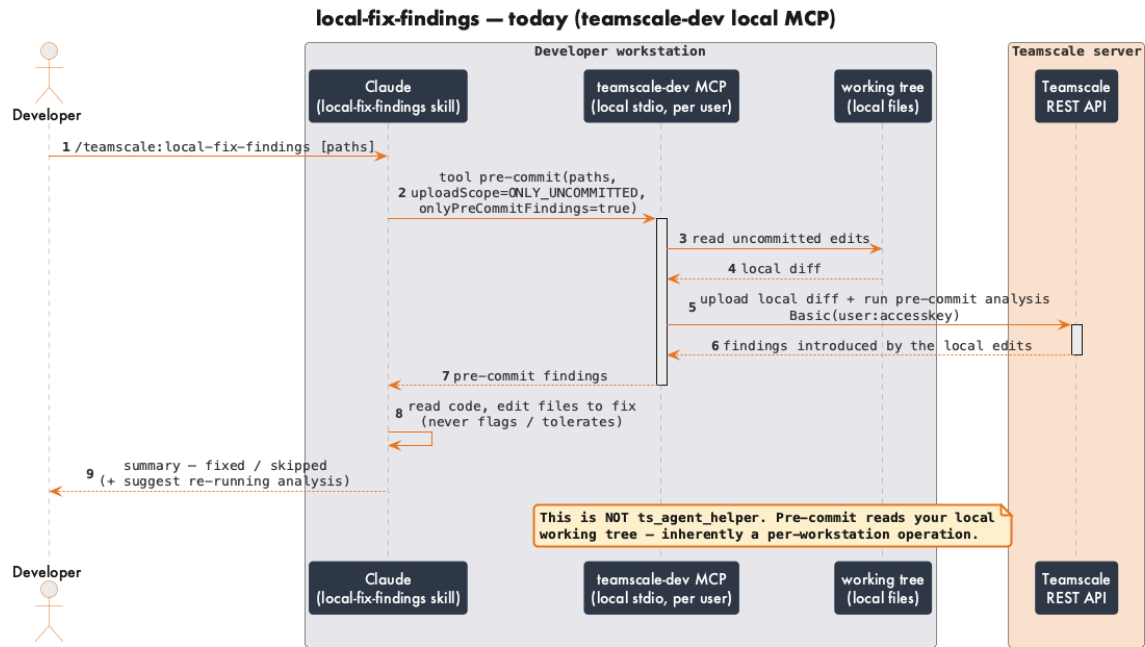


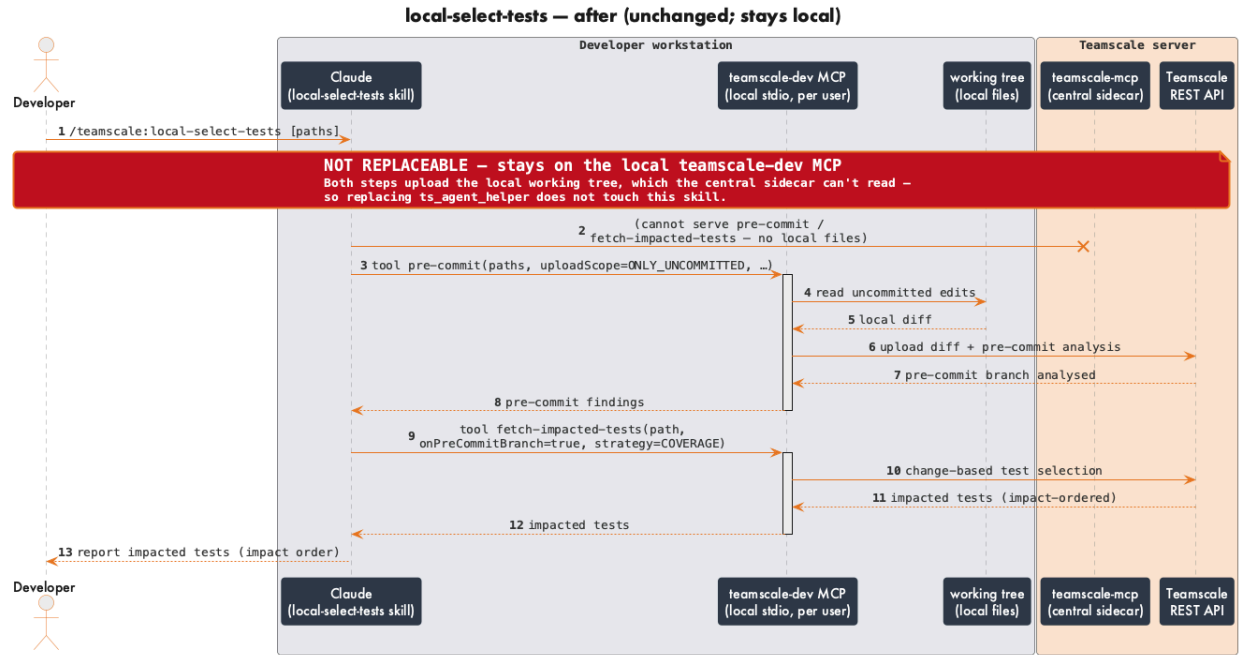
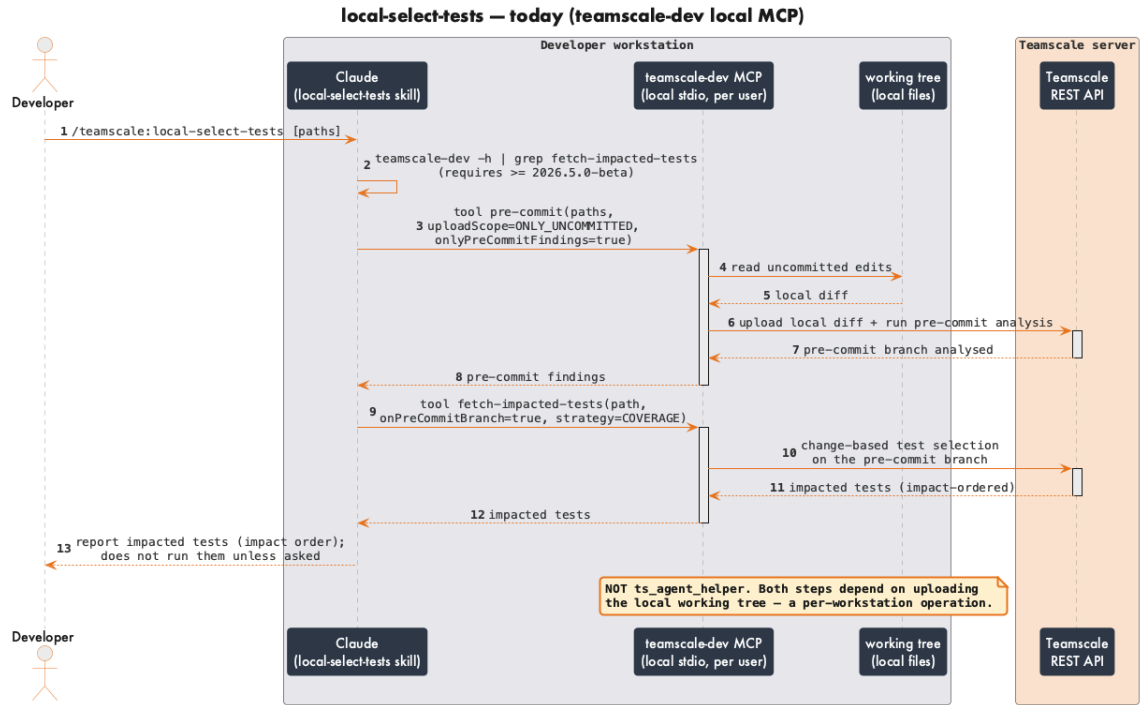
Plugin comparison

Each diagram: **today** (`ts_agent_helper`) on the left, **with** `teamscale-mcp` on the right.

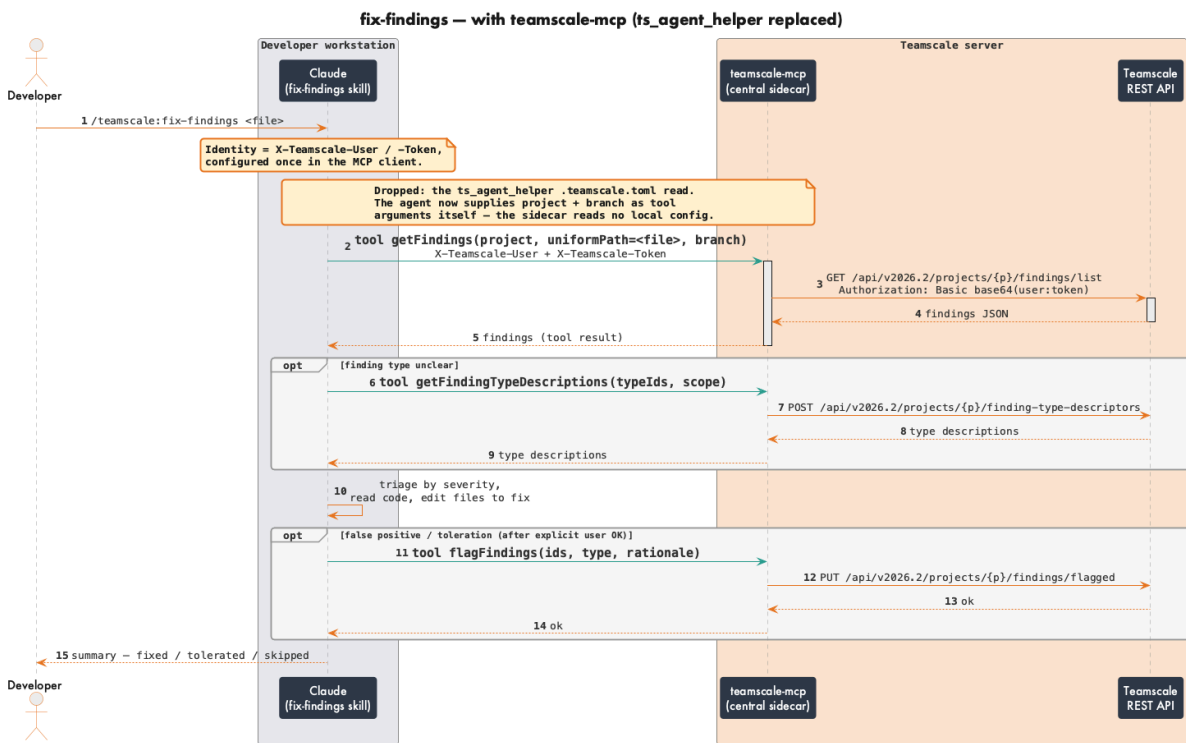
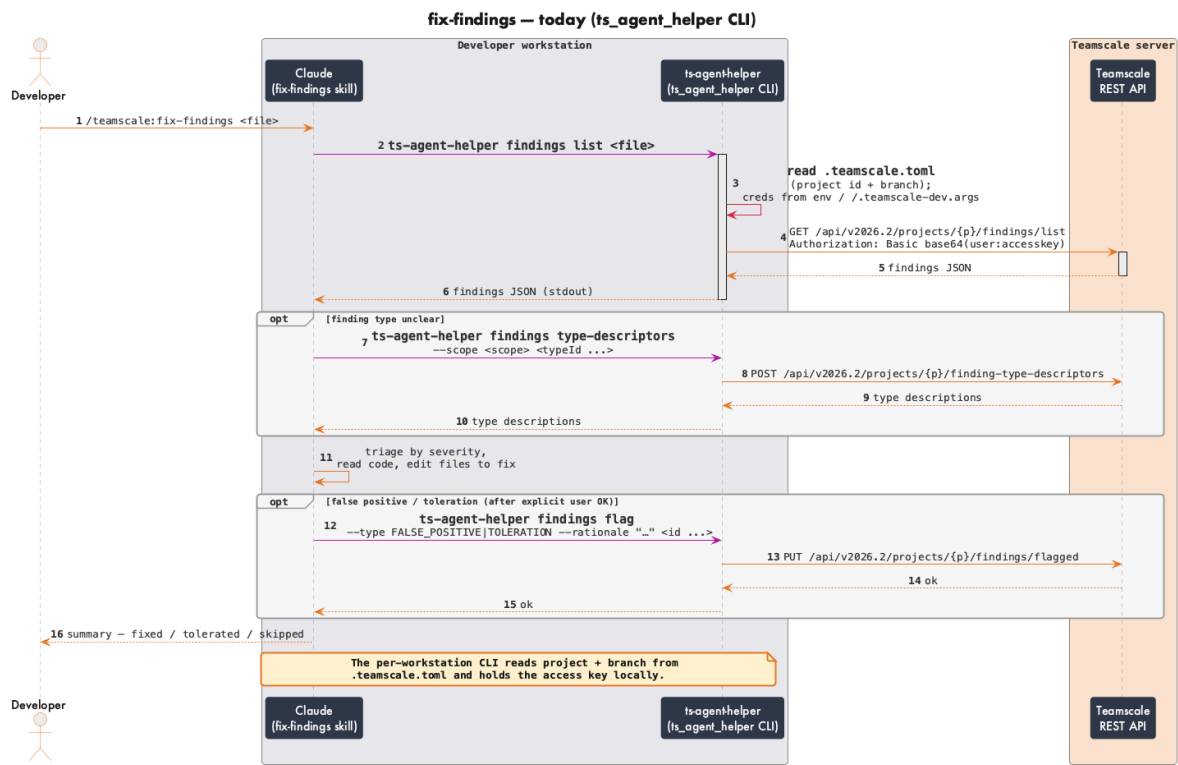
check-setup — shifts to verifying the endpoint



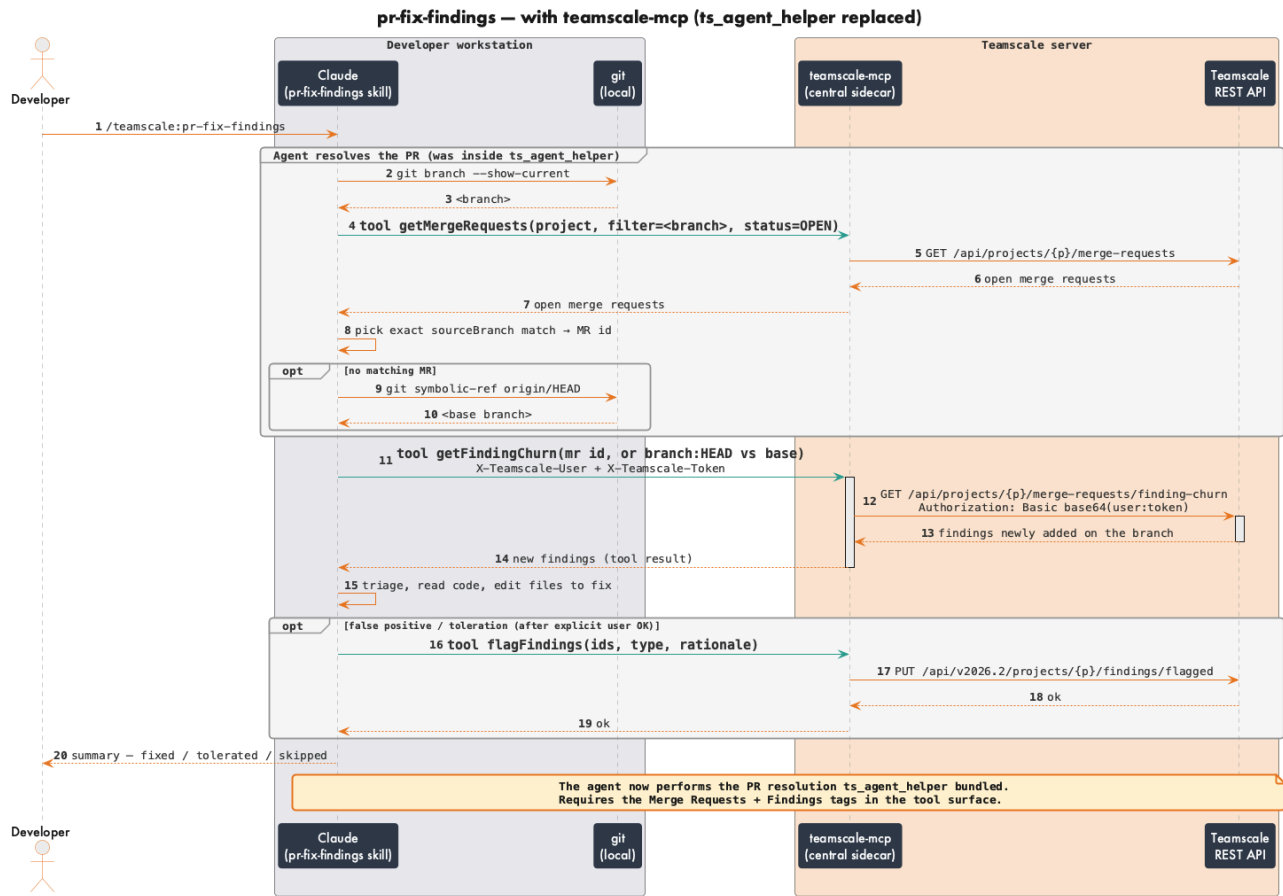
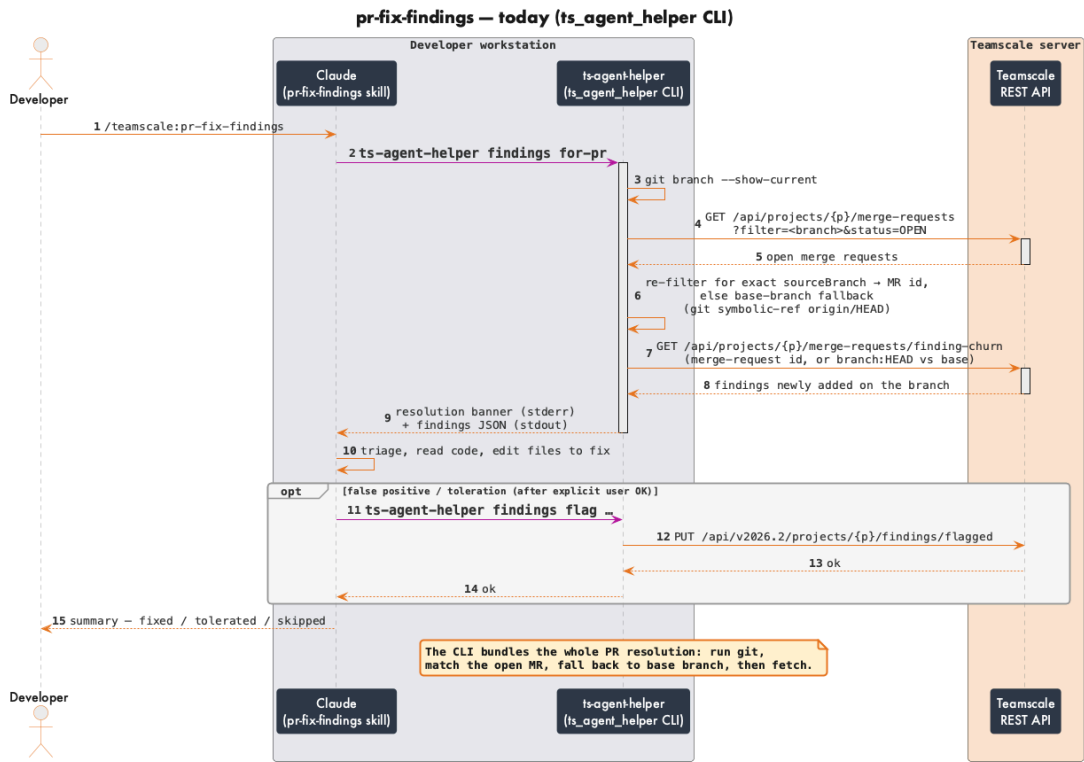




fix-findings — replaceable (clean win)



pr-fix-findings — replaceable (agent takes over PR resolution)



pr-close-test-gaps — replaceable (agent takes over PR resolution)

